INCH-POUND

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SUPERSEDING
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MILITARY SPECIFICATION

TANK, FABRIC, COLLAPSIBLE; AIR COLUMN SUPPORTED, OPEN TOP, WATER STORAGE 3000 GALLONS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This document covers a fabric, collapsible, air column self-supporting, 3000 gallon tank for storage of drinking water.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5430

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SPECIFICATIONS

FEDERAL

A-A-55057	- Panel Wood/Wood Based, Construction and Decorative.
L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
FF-B-561	- Bolt, Lag.
FF-B-584	- Bolt, Finned Neck, Key Head, Machine, Ribbed Neck, Square Neck, Tee Head.
FF-N-836	- Nut, Square, Hexagon Cap, Slotted, Castellated, Clinch, Knurled, and Welding.
FF-W-92	- Washer, Metal, Flat (Plain).
TT-W-572	- Wood Preservative; Water Repellant.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-C-795	- Cushioning Material, Flexible, Cellular, Plastic Film, for Packaging Applications.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular Polypropylene Foam.
PPP-T-60	- Tape, Packaging, Waterproof.
PPP-T-97	- Tape, Pressure Sensitive Adhesive, Filament Reinforced.

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MIL-P-116	- Preservation, Methods of.
MIL-T-22085	- Tape, Pressure Sensitive, Adhesive, Preservation and Sealing.
MIL-R-52255	- Repair Kit and Repair Kit Components for Collapsible Fabric Tanks and Boats.

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Method.
FED-STD-313	- Material Safety Data Sheet.
FED-STD-595	- Color.
FED-STD-601	- Rubber: Sampling and Testing.

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-731	- Quality of Members for Containers and Pallets.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

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MS27023	- Coupling Half, Quick Disconnect, Cam-Locking Type Male, Flanged, Type IV.
MS27027	- Coupling Half, Quick Disconnect, Cam-Locking Type Female, Flanged, Type IV.
MS27028	- Coupling Half, Quick Disconnect, Cam-Locking Type Cap, Dust, Type IX.
MS27029	- Coupling Half, Quick Disconnect, Cam-Locking Type Plug Dust, Type X.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in the solicitation.

U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

FOOD AND DRUG ADMINISTRATION (FDA)

21 CFR - Food and Drugs, Parts 170-199

(Application for copies should be addressed to the Super-intendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues cited in the solicitation (see 6.2).

AMERICAN PUBLIC HEALTH ASSOCIATION, INC. (APHA)

Standard Methods for the Examination of Water and Waste Water

Part 207 - Odor.
Part 211B - Taste.

(Application for copies should be addressed to the American Public Health Association, Inc., 1015 18th St. NW, Washington, DC 20036.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 413	-	Rubber Properties - Adhesion to Flexible Substrate.
D 429	-	Rubber Property - Adhesion to Rigid Substrates.
D 471	-	Rubber Property - Effect of Liquids.
D 1149	-	Rubber Deterioration - Surface Ozone Cracking in a Chamber.

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D 2261 - Tearing Strength of Woven Fabrics by the Tongue Method.
D 2565 - Operating Xenon Arc - Type (Watercooled) Light-Exposure Apparatus With and Without Water for Exposure of Plastics.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WOOD PRESERVERS' ASSOCIATION (AWPA)

C1 - All Timber Products - Preservative Treatment by Pressure Process.
C2 - Lumber, Timbers, Bridge Ties and Mine Ties Preservative Treatment by Pressure Process.
C9 - Plywood - Preservation Treatment by Pressure Process.

(Application for copies should be addressed to the American Wood Preservers' Association, P.O. Box 849, Stevensville, MD 21666.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard 61 - Drinking Water System Components - Health Effects.

(Application for copies should be addressed to the National Sanitation Foundation, 3475 Plymouth Rd., P.O. Box 1468, Ann Arbor, MI 48106.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J105 - Hex Bolts.

(Applications for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The tank shall be capable of holding 3000 gallons of water. The upper portion of the 3000 gallon open-top storage tank shall consist of a circular collar which can be inflated by means of a foot-type inflation bellows, which shall be provided with the tank. A closely-fitted protective cover shall be provided for the tank. The protective cover shall

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fasten to the tank by plastic toggles and loops, and shall be supported by an inflatable ball or pillow which is attached to the underside of the cover. The tank shall be equipped with one male and one female standard quick connect fitting as specified in 3.7.3. The fittings shall be located 180 degrees apart near the bottom of the tank such that the fittings are accessible when the tank is filled to rated capacity. A reinforcing pad shall be bonded to the tank surrounding the fitting connections to support the fitting to the tank fabric interface. A fabric valise shall be provided for carrying the tank and accessories. The valise may be the cover or ground cloth which is fitted with handles and closures (i.e. snaps) and folding instructions for use as the valise. The total weight shall not exceed 135 pounds including the tank, the cover with float, the valise, the inflation bellows, the repair kits and the instruction manual; but does not include the ground cloth. No part of the tank interior, cover, or float shall have a detrimental effect on potable water. The tank, cover, float, ground cloth, fittings, and valise shall approximate color 33446 or 37038 of FED-STD-595 (see 6.2g).

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.4). The sample shall consist of 2 units (see 6.4).

3.3 Sample selection plan. Prior to initial production, the contractor shall have a sample selection plan (see 6.2). The plan shall be designed for use in the quality conformance testing of section 4 and shall provide the Government a method of selecting production seams, bonded fitting, coated fabric and handle samples for destructive tests. The samples shall be cut from a production tank selected at random in accordance with 4.4.2.1. The test specimens of seams, bonded fittings, coated fabrics and handles shall be provided in sufficient quantity for testing as specified herein. The sample selection plan shall be made available, upon request, to the contracting officer or Government representative on site of manufacturer.

3.4 Samples for submission. The contractor shall furnish within 30 days of commencing first article testing, samples of coated fabric (one square yard) and of seam (36 inches of seam with 6 inches minimum of coated fabric on either side.) The samples shall be sent to Belvoir Research, Development and Engineering Center, ATTN: STRBE-FSH, Ft Belvoir, VA 22060-5606. Samples shall be tagged with the appropriate contract number.

3.5 Process data. Within the time frame specified (see 6.2), the contractor shall make available to the US Government representative on site of manufacturer, a description of materials and documented inspection procedures which will be used by the contractor to ensure that each production tank is a duplicate of the first article. Change of materials, coated fabric, components, procedures, or sources of supply from those initially selected shall be subject to the approval of the contracting officer (see 6.3 and 6.6) and shall require that examination and testing be performed as referenced in 3.2.

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3.6 Material. Material shall be as specified herein and as described by tables I, II, and III herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification. The finished fabric tank shall contain no materials or substances which might leach out or deteriorate and cause the water to become nonpotable. The material shall have no adverse effect on the health of personnel when used for its intended purpose. All surfaces which come in contact with potable water shall be in accordance with the Code of Federal Regulations (CFR), Title 21 - Food and Drugs, Part 177, less sections 177.1020 through 177.1050 and 177.1480 or shall be tested or listed as approved for use with potable water by the National Sanitation Foundation (NSF) Underwriters Laboratory, or the Safe Water Additives Institute in accordance with NSF Standard 61.

3.6.1 Material deterioration prevention and control. The tank shall be fabricated from compatible materials, inherently deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environments to which the tank may be exposed.

3.6.2 Identification of materials and finishes. The contractor shall identify the specific materials, material finishes and treatments for use with components and subcomponents, and shall make information available upon request to the contracting officer or his designated representative.

3.6.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the tank may be newly fabricated from recovered materials to the maximum extent practicable, provided the tank produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the tanks.

3.6.4 Cloth. The cloth of the coated fabric shall be high-tenacity, and heat and light resistant. The cloth shall be free from any imperfection affecting strength, coating adhesion, coating thickness, or leakage. Cloth shall be nylon or polyester.

3.6.5 Coating compounds. Every compound used for the formulation of elastomeric coatings, tie gums, diffusion barrier (if used), and any other materials applied to the fabric and subject to contact with drinking water shall be suitable for use with drinking water (see 3.6). All coating compounds shall be suitable for use when in continuous contact with rainwater, ground water, or well water and shall be fungus and weather resistant.

3.6.6 Coated fabric. The coated fabric shall be free from blisters or pinholes and shall show no signs of coating delamination. The coated fabric shall withstand the effects of humidity, ozone, and weather elements without damage, deterioration, or failure to meet performance requirements specified herein. The coated fabric shall conform to the requirements of table I. The

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protective cover and float are excluded from the requirements of table I, item 1 through 4.

TABLE I. Characteristics of coated fabric.

Item	Property	Requirements	Test paragraph, ASTM test method or test method of FED-STD-191
1.	Tearing strength warp and fill.	26 lbs (min)	4.5.1.10/ D 2261
2.	Breaking strength	175 lbs/in (min)	4.5.1.10/5102
3.	Weathering resistance after 1500 hours exposure at 5 percent elongation, warp and fill (% retention of breaking strength).	80% (min)	4.5.1.10/5804 1/ or D 2565 2/ 5102
4.	Puncture resistance.	110 lbs (min)	4.5.1.11/5120
5.	Low temperature crease resistance (appearing after unfolding).	No cracking, peeling or delamination	4.5.1.12
6.	Blocking.	Separation of specimens to occur within 5 seconds	4.5.1.13
7.	Coating adhesion, initial.	20 lbs/in (min)	4.5.1.14
8.	Coating adhesion at 73 ±5 °F after immersion in distilled water at 160 ±2 °F for the following durations		
	14 days	15 lbs/in (min)	4.5.1.14
	42 days	12 lbs/in (min)	4.5.1.14
9.	Ozone resistance	No cracks under 7X lens	4.5.1.21 D 1149

1/ Specimens shall have exterior coating facing carbon arc. Alternative Corex D filters shall be removed.

2/ Xenon light, procedure A, inner and outer borosilicate filters, deionized water (68 ±5 °F); 690 minutes light exposure, 30 minutes light and gray, block panel temperature (145 ±5 °F); relative humidity (45 ±5 percent). Coated fabric specimens shall have exterior coating (outside of tank) facing the light.

3.7 Tank construction. The tank shall be fabricated from coated fabric as specified herein. The configuration and dimensions of the tank, cover, and float and location of the fittings, labels, and handles shall be as shown in figures 1 and 2. Seams shall not coincide with tank fittings. For both tank

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interior and exterior, all coated fabric edges (seams, chafing patches, fabric flanges of fittings) shall be covered with coated fabric or coating compound used to fabricate the tank. The inflation collar and float shall be capable of preventing a pressure drop exceeding 5 inches of water in 60 minutes when tested as specified in 4.5.1.2. The inflation collar and float shall be fitted with one inflation and deflation port. These ports shall be labeled with curved double-ended arrows with the words "INFLATION" and "DEFLATION" placed appropriately to indicate operation of the port valves. The inflation collar and float shall be fitted with one pressurized air inflation valve (automotive type) labeled "INFLATION". These labels shall be stencilled in 1-inch high letters and shall be in accordance with 3.18. The tank shall not leak. The inflation collar and float shall not leak.

3.7.1 Seams. All tank seams including end closures, handle patches, and fabric flanges of fittings shall conform to the requirements of table II. Seams between adjacent panels of coated fabric shall be constructed so as to prevent wicking of water through the coated fabric.

TABLE II. Characteristics of seams.

Item	Property	Requirements	Test paragraph, test methods of FED-STD-601 or ASTM test method
1.	Breaking strength (initial).	175 lbs/in (min)	4.5.1.15/8311
2.	Breaking strength at 73 ±5 °F after immersion in distilled water at 160 ±2 °F for the following durations:		
	14 days	170 lbs/in	4.5.1.15/8311/ 6001
	42 days	160 lbs/in	4.5.1.15/8311/ 6001
3.	Dead load shear resistance under 50 lbs/in stress at 160 ±2 °F for 8 hours.	0.1 inch slippage (max)	4.5.1.16
4.	Peel adhesion.	20 lbs/in (min)	4.5.1.15/D 413 (Machine Method)
5.	Peel adhesion at 73 ±5 °F after immersion in distilled water at 160 ±2 °F for the following durations:		
	14 days	15 lbs/in (min)	4.5.1.15/D 413 (Machine Method)
	42 days	12 lbs/in (min)	
6.	Handle seams pull test 1/	135 lbs	4.5.1.6

1/ Applies to seams between handles and tank, and handles and valise.

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3.7.2 Handles. The handles shall be U-shaped and fabricated from nylon webbing or cloth and fastened to a coated fabric patch of the same material as the tank. The handles shall be bonded to the tank wall such that they are at the side of the tank (not under) when the tank is filled to its rated capacity. Ten handles shall be spaced equidistantly from one another. A handle shall be provided for the inside, center floor of the tank for suspending the tank inside out. The bonds between each handle patch assembly and the tank fabric shall conform to the requirements of table II.

3.7.3 Fittings. The tank shall be furnished with one male and one female filler/discharge assembly, positioned at ground level and 180 degrees from one another. The male filler/discharge assembly shall be in accordance with MS27023-11 with a cap in accordance with MS27028-11 and chain. The female filler/discharge assembly shall be in accordance with MS27027-11 with plug in accordance with MS27029-11 and a chain. Above each filler/discharge assembly the tank shall be labeled in 0.50-inch black letters "CONNECT FILLER/ DISCHARGE HOSE HERE".

3.7.3.1 Reinforcing pad assembly. A reinforcing pad assembly shall be bonded into the coated fabric to support the filler/discharge assemblies. The bond between the reinforcement pad and the tank wall shall meet the requirements presented in table III.

TABLE III. Characteristics of bonded reinforcing pad.

Item	Property	Requirements	Test paragraph, or ASTM test method
1.	Reinforcing pad to coated fabric bond strength.	180 lbs/in (min)	4.5.1.17
2.	Bond strength at 73 ±5 °F after immersion in distilled water at 160 ±2 °F for the following durations:		
	14 days	160 lbs/in (min)	4.5.1.17
	42 days	150 lbs/in (min)	4.5.1.17
3.	Dead load shear resistance under 50 lbs/in stress at 160 ±2 °F for 8 hours.	0.1 inch slippage (max)	4.5.1.17
4.	Peel adhesion of reinforcing pad to coated fabric.	15 lbs/in (min)	4.5.1.18
5.	Peel adhesion at 75 ±5 °F after immersion in distilled water at 160 ±2 °F for the following durations:		
	14 days	11 lbs/in (min)	4.5.1.18/D 429 (Method B)
	42 days	9 lbs/in (min)	

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3.7.3.2 Bolts. The fittings shall be bolted to the bonded reinforcing pad using bolts, hex head, 1/4-20 UNC, steel, zinc-coated, 0.003 inches thick, in accordance with SAE J105.

3.8 Tank performance. The tank and components shall withstand folded storage at ambient temperatures from -25° to 160 °F without damage or leakage when subsequently filled with water. The tank and components shall be suitable for operational use at ambient temperatures from 32° to 125° F. The tank shall not be damaged when exposed to relative humidity up to 100 percent. The tank shall be suitable for use in continuous contact with rainwater, well water, and ground water. There shall be no evidence of leakage or seepage when the tank is filled to its rated capacity with water (see 4.5.1.5). The tank shall be capable of being filled on level ground without any support that is not integral to the tank. The tank, valise, repair kits, cover, float, bellows, and instruction manual shall have a total weight of not more than 135 pounds.

3.9 Storage/carrying valise. The tank shall be furnished with a fabric valise. The valise shall be used to carry the tank, inflation bellows, repair kit, protective cover, float, ground cloth, and instruction manual. An encapsulated plastic instruction card shall be attached to the inside valise flap, stating set-up and repacking instructions. Any breaks in stitching of seams of the valise shall be overstitched not less than 2 inches at each break. Stitching shall not be loose. The valise shall have a service and storage life equal to that of the tank. The valise shall be provided with the appropriate amount of handles to permit a four-soldier carry and shall be labeled with "4 Soldier Carry" in one-inch high uppercase letters, in accordance with 3.18. The fully loaded valise shall be capable of withstanding being dropped from a height of 60 inches onto a hard surface without failure when tested as specified in 4.5.1.9. A failure shall be defined as a rip, tear, split, or break in the fabric or seams of the valise or damage to its contents. The bonds between each handle patch assembly and the valise fabric shall conform to table II requirements.

3.10 Medical standards.

3.10.1 Taste and odor. The materials of the tank to contact drinking water, float repair kit material, and cover material shall conform to APHA Standards and Methods for Examination of Water and Waste-water, parts for Taste (211B) and Odor (207). The material shall not impart odor to chlorinated water such that the threshold odor number exceeds 2. The material shall not impart taste to chlorinated water such that the taste rating scale exceeds 4.

3.11 Repair kits. An adhesive repair kit shall be provided with the tank. The repair kit shall contain adhesives and patches sufficient to perform field repair on four 4-inch slits in the tank wall (see 4.5.1.5.2). Containers for adhesives shall be labeled with contents and any appropriate warnings. Repairs accomplished with the repair kit shall conform to the standards herein (see 3.10.1). In addition, a repair kit conforming to MIL-R-52255 type II

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shall be provided with the tank. Both repair kits shall be contained in a pouch on the side of the tank. The pouch shall be permanently marked "Repair Kits" in 2-inch letters.

3.11.1 Material safety data sheet. Material Safety Data Sheets shall be prepared in accordance with FED-STD-313 (see 6.12).

3.12 Inflation bellows. A foot-type inflation bellows shall be provided with the tank to be used to inflate the tank collar and float. The inflation bellows shall be capable of producing a minimum of 3 psi per cycle over 10,000 cycles when tested as specified in 4.5.1.8.

3.13 Protective cover and float. A protective cover and inflatable float shall be provided with the tank. The inflatable float shall be capable of being attached and removed from the cover. The cover shall have nylon webbing loops on its perimeter to secure the cover to the tank. The cover shall conform to figure 2. The cover and float shall be of coated fabric as specified in table I. The float shall be spherical or pillow shaped and shall be 24 inches (min) height or diameter.

3.14 Ground cloth. A ground cloth shall be furnished with each tank. The ground cloth shall extend 6 inches beyond the edge of the tank when it is filled to its rated capacity. The ground cloth shall meet the requirements of the coated fabric used in the construction of the tank (see 3.6.6). All exposed edges of coated fabric shall be sealed.

3.15 Color. The tank, cover, float, ground cloth, fittings, and valise shall approximate color 33446 or 37038 of FED-STD-595 (see 6.2).

3.16 Identification marking.

3.16.1 Tank and valise. The tank and valise shall be identified in accordance with MIL-STD-130 by means of an identification label. The following information shall be provided using 0.50-inch (minimum) letters on each tank and valise.

TANK, FABRIC, SELF-SUPPORTING
3000 U S GALLON WATER
NSN: (Specify)
SERIAL NO: (Specify)
MFR: (Manufacturer's name and location of plant)
WEIGHT EMPTY: (Specify number of pounds)
CONTRACT OR ORDER NO: (Specify)
LOT: (Specify)
DATE OF MANUFACTURE: (Specify month and year)
FIVE YEAR USE LIFE - EXTENDABLE
TEN YEAR STORAGE LIFE - EXTENDABLE

Identification labels shall conform to 3.18:

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3.16.2 Cover. The cover shall be permanently marked in black 4-inch lettering "DRINKING WATER ONLY". All labels shall conform to 3.18.

3.17 Workmanship. Workmanship shall be of the highest quality and shall permit no defects not repaired or reworked to full integrity in accordance with accepted process data and 3.20. All metal parts shall be clean and free of sand, dirt, scale, and flux. Surfaces shall be smooth with edges rounded or beveled. The inside and outside of the tank shall be clean and free of foreign materials. Any necessary repair or rework shall restore the reworked area to its full strength and shall meet all applicable requirements of this specification. The cemented surfaces of all spliced areas, fitting flanges, and patch-type repairs shall effect a bond that will result in strength of the cemented area not less than the strength of adjacent tank fabric.

3.18 Human factors engineering. The tank, valise, float, protective cover, inflation bellows, and repair kit shall be designed in accordance with MIL-STD-1472, valve controls (5.4.1.2.4), labeling (5.5.1), handles and grasp areas (5.9.11.5).

3.19 Stability. The water filled tank shall be usable on slopes up to 10 percent (10 feet per 100 feet). When tested in accordance with 4.5.1.19, the tank shall not collapse, creep, or overturn.

3.20 Rework and repair. Any rework or repair of seams shall not exceed 5 percent of the total seam length of the tank. Total repair or rework on any seam shall not exceed the length of 5 percent of the length of that seam. Repair and rework of areas of the tank body fabric shall be limited to areas of damage of 3 inches (length, width, diameter) or less. All rework and repair procedures must be accomplished before quality conformance inspection and inspection of packing. The contractor shall make available to the contracting officer and Government representative on site of manufacture a rework and repair procedure for use during production of the tanks (see 6.2). Any deviation from the procedure shall require the prior approval of the contracting officer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality

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program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.3 Process inspection. Inspection shall be accomplished by the contractor to assure compliance with the approved documented work instruction covering processing and fabricating procedures. Any process or fabricating procedure not in accordance with the approved procedure shall be cause to withhold acceptance of tanks determined to have been manufactured using the unapproved process or fabricating procedure.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article. Unless otherwise specified (see 6.2), the first article inspection will be performed by the contractor. Acceptance of the first article tanks shall not exclude the remaining tanks from meeting all requirements of this specification. In addition to the performance of first article inspections, the Government reserves the right to perform any and all inspections contained in this specification to assure compliance. Failure to meet any requirement of this specification shall constitute failure of first article inspection.

4.3.1 First article examination. The first article tank(s) shall be examined as specified in table IV. Presence of any defect shall be cause for rejection.

TABLE IV. Examination schedule.

Examination number	Examination description	Requirement paragraph
101.	Duplicate samples not sent to STRBE-FSH, Belvoir Research, Development and Engineering Center.	3.4
102.	Process data not available for review.	3.5
103.	Changed process data not approved by the contracting officer.	3.5
104.	Materials not as specified.	3.6

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TABLE IV. Examination schedule - continued.

Examination number	Examination description	Requirement paragraph
105.	Materials are not resistant to deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating operating environments.	3.6.1
106.	Contractor does not have documentation available for identification of material, material finishes, or treatment.	3.6.2
107.	Used, rebuilt, or remanufactured components, pieces or parts incorporated in the tanks.	3.6.3
108.	Cloth not as specified.	3.6.4
109.	Coating compound not as specified.	3.6.5
110.	Blisters or pinholes in coated fabric.	3.6.6
111.	Dimensions not as specified.	3.7, 3.13 3.14
112.	Edges of coated fabric not covered as specified.	3.7
113.	Pressurized air inflation valve not as specified.	3.7
114.	Handles not as specified.	3.7.2
115.	Fittings not as specified.	3.7.3
116.	Reinforcing pad not as specified.	3.7.3.1
117.	Bolts not as specified.	3.7.3.2
118.	Valise not as specified.	3.9
119.	Valise stitching not as specified.	3.9
120.	Repair items missing or not as specified.	3.11
121.	Material Safety Data Sheet not as specified	3.11.1
122.	Inflation bellows not as specified.	3.12
123.	Protective cover and float not as specified.	3.13
124.	Ground cloth size not as specified.	3.14
125.	Tank, cover, float, valise, ground cloth, and fittings color not as specified.	3.15
126.	Identification label not as specified.	3.16, 3.18
127.	Identification marking missing, incomplete, or illegible.	3.16, 3.18
128.	Extraneous material inside tank.	3.17
129.	Workmanship not as specified.	3.17
130.	Human factors engineering not as specified.	3.18
131.	Rework exceeds limits specified.	3.20

4.3.2 First article tests. The first article tank(s) shall be subject to tests marked "X" in column I of table V. Failure of any test shall be cause for rejection.

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TABLE V. Test schedule.

First article	Quality conformance: Individual	Quality conformance: Samples	Test category	Inspection procedure	Requirement
1	2	3	4	5	6
			Tank and accessories		
X	-	X	Weight & dimension.	4.5.1.1	3.7, 3.8
X	X	-	Air-leakage.	4.5.1.2	3.7
X	-	-	Low temperature.	4.5.1.3	3.8
X	-	-	Taste and Odor.	4.5.1.20	3.10.1
X	-	-	High temperature.	4.5.1.4	3.8
X	-	-	Water storage.	4.5.1.5	3.8
-	-	X	Initial water storage.	4.5.1.5.1	3.8
X	-	X	Handle pull test.	4.5.1.6	3.7.2
X	X	-	Internal inspection.	4.5.1.7	3.17
X	-	-	Inflation bellows.	4.5.1.8	3.12
X	-	X	Storage/carrying valise.	4.5.1.9	3.9
X	-	X	Stability.	4.5.1.19	3.19
X	-	-	Adhesive repair.	4.5.1.5.2	3.11
			Coated fabric		
X	-	X	Tearing strength.	4.5.1.10	3.6.6, table I
X	-	X	Breaking strength.	4.5.1.10	3.6.6, table I
X	-	-	Weathering resistance.	4.5.1.10	3.6.6,
X	-	X	Puncture resistance.	4.5.1.11	3.6.6, table I
X	-	-	Low temperature crease resistance.	4.5.1.12	3.6.6, table I
X	-	X	Blocking.	4.5.1.13	3.6.6, table I
X		X	Coating adhesion initial.	4.5.1.14	3.6.6, table I
X	-	X	Coating adhesion after 14 day water immersion.	4.5.1.14	3.6.6, table I
X	-	-	Coating adhesion after 42 day water immersion.	4.5.1.14	3.6.6, table I
X	-	-	Ozone resistance.	4.5.1.21	3.6.6, table I

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TABLE V. Test schedule - continued.

First article	Quality conformance: Individual	Quality conformance: Samples	Test category	Inspection procedure	Requirement
1	2	3	4	5	6
			Seams		
X	-	X	Breaking strength (initial).	4.5.1.15	3.7.1, table II
X	-	X	Breaking strength after 14 day water immersion.	4.5.1.15	3.7.1, table II
X	-	-	Breaking strength after 42 day water immersion.	4.5.1.15	3.7.1, table II
X	-	X	Dead load shear resistance.	4.5.1.16	3.7.1, table II
X	-	X	Peel adhesion (initial).	4.5.1.15	3.7.1, table II
X	-	X	Peel adhesion after 14 day water immersion.	4.5.1.15	3.7.1, table II
X	-	-	Peel adhesion after 42 day water immersion.	4.5.1.15	3.7.1, table II
			Bonded reinforcing pad		
X	-	X	Bond strength (initial).	4.5.1.17	3.7.3, table III
X	-	X	Bond strength after 14 day water immersion.	4.5.1.17	3.7.3, table III
X	-	-	Bond strength after 42 day water immersion.	4.5.1.17	3.7.3, table III
X	-	X	Dead load shear resistance.	4.5.1.17	3.7.3, table III
X	-	X	Peel adhesion (initial).	4.5.1.18	3.7.3, table III
X	-	X	Peel adhesion after 14 day water immersion.	4.5.1.18	3.7.3, table III
X	-	-	Peel adhesion after 42 day water immersion.	4.5.1.18	3.7.3, table III

4.4 Quality conformance inspection.

4.4.1 Individual tanks. Each complete tank assembly shall be examined as specified in table IV, and subjected to the tests marked "X" in column 2 of

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table V. Any nonconformance revealed by the examination or failure of the test shall be cause for rejection of the tank.

4.4.2 Sample testing.

4.4.2.1 Sampling for destructive test. Selection of samples for destructive tests shall be in accordance with the sample selection plan (see 3.3). The frequency for random selection of production tanks will be once for each 26 tanks produced. The 26 tanks will be identified as a lot corresponding to the selected samples. The tank used for destructive testing shall remain the property of the contractor. The remaining 25 tanks are deliverable tanks in that lot. The samples used for destructive testing shall remain the property of the contractor. Continuous successful completion of four test cycles (see 4.4.2.2) may be cause for reducing the frequency of testing, subject to approval of the contracting officer. Any reduction in frequency will cause a corresponding increase in the represented lot size. Rejection of any lot (see 4.4.2.2) during the reduced selection frequency will require a return to the original frequency.

4.4.2.2 Testing. Samples selected in accordance with 4.4.2.1 shall be subjected to the tests marked "X" in column 3 of table V. Failure of any test shall be cause for rejection of the represented lot of tanks.

4.5 Inspection procedure.

4.5.1 Tests.

4.5.1.1 Tank and accessory weight. Weigh the tank, valise, repair kits, cover, float, instruction manual, and bellows for compliance with 3.8. Inability to meet the weight requirement shall constitute failure of this test.

4.5.1.2 Air-leakage. The inflation collar and float shall be initially inflated to a pressure of 60 inches of water for 30 minutes. The inflation collar and float shall then be inflated to a pressure of 55 inches of water for 60 minutes. The tank shall be set up in an area shielded from direct sunlight with temperatures and time recorded at start of 60 minute test phase. A variation in temperature during the 60 minute test phase of more than 3.6 °F shall cause the test to be repeated. At the end of the 60 minute test phase, the time, temperature and pressure shall be recorded. A correction in temperature shall be made for each degree F, as follows:

a. For a rise in temperature, subtract 1.5-inches of water per 1.8 °F.
b. For a fall in temperature, add 1.5-inches of water per 1.8 °F.

A drop in pressure of more than 5 inches of water after temperature correction shall constitute failure of this test.

4.5.1.3 Low temperature. The tank and accessories shall be placed in the storage/carrying valise and subjected to a low temperature environment of

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-28 ±2 °F, for a period of 24 hours. The environmental temperature shall then be increased to -25 °F, for an additional 24 hours. At the end of this period, while still at -25 °F, the tank shall be slowly unfolded in not less than 15 minutes nor more than 30 minutes. The tank shall be removed from the low temperature environment. The inflation bellows shall be used for inflating the collar. The tank shall be filled with 3000 gallons of water. Any flaking, cracking, or separation of the coated fabric, failure of the inflation bellows, seepage or leakage shall constitute failure of this test.

4.5.1.4 High temperature. The tank and accessories shall be placed in the storage/carrying valise and subjected to a high temperature environment of 160 ±2 °F, for a period of 24 hours. The environmental temperature shall then be decreased to 125 ±2 °F for an additional 24 hours. At the end of this period, while still at 125 ±2 °F, the tank shall be slowly unfolded in not less than 15 minutes and not more than 30 minutes. The tank shall be removed from the high temperature environment. The inflation bellows shall be used for inflating the collar. The tank shall be filled with 3000 gallons of water. Any flaking, cracking, or separation of the coated fabric, failure of the inflation bellows, seepage or leakage shall constitute failure of this test.

4.5.1.5 Water storage.

4.5.1.5.1 Initial water storage. The tank shall be set up indoors with cover and float, filled with 3000 gallons of potable water through a filler/discharge assembly, allowed to stand for 24 hours, and examined for seepage or leakage. Any evidence of seepage or leakage or failure of tank to contain 3000 gallons shall constitute failure of this test.

4.5.1.5.2 Water storage after repair. Four repairs of 4-inch slits shall be made in the tank wall using the adhesive repair kit. The tank shall be set up indoors with the cover and float, filled with 3000 gallons of potable water, allowed to stand for 24 hours and examined for seepage and leakage around the patched area. Any evidence of seepage or leakage shall constitute failure of this test.

4.5.1.6 Tank handle pull resistance. The test item shall consist of four handles, which are spaced evenly around the tank, as well as a suspension handle attached to the inside bottom of the tank. Each handle shall be tested in accordance with the following procedure. The body fabric shall be tightly drawn and clamped between two flat oval rings or a base plate and an oval ring so that the handle of the patch is centrally located in the rings. The shape of the oval rings shall be such that all edges of the coated fabric patch are 3 inches, ±1 inch, from the ring clamp. The rigidity, strength, and construction of the clamp shall be such that the tank body material shall not slip more than 0.50 inch at any point during the test. With the sample held securely, a tension shall be applied through bar or pipe, 1 inch in diameter, inserted in the loop of the handle. The tension shall be applied slowly and smoothly perpendicular to the plane of the handle patch until the load of 500 pounds is reached. The 500 pound load shall be maintained for 1 minute. Any

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damage, permanent distortion, or separation of the handle patch, or tank material shall constitute failure of the test.

4.5.1.7 Internal inspection. The tank shall be inspected internally. Any evidence of weakened areas, coating, or barrier delamination shall constitute failure of this test.

4.5.1.8 Inflation bellows. The inflation bellows shall be cycled, 20 times per minute, against a static head of 3 psi (minimum) for 10,000 cycles. Inability of the inflation bellows to develop a pressure of at least 3 psi per cycle over 10,000 cycles shall constitute failure of this test. The inflation bellows shall be used to inflate the collar to 3 psi (minimum) after 120 cycles (maximum). Inability to achieve 3 psi (minimum) after 120 cycles (maximum) shall constitute failure of this test.

4.5.1.9 Storage/carrying valise. The storage/carrying valise loaded with tank and all accessories shall be dropped from a height of 60 ±2 inches. The valise and its contents shall be examined for failure. A failure is defined as a rip, tear, split or break in the fabric or seams or damage to contents. Three valise handles shall be tested using procedure of 4.5.1.6. Nonconformance to 3.9 shall constitute failure of this test.

4.5.1.10 Coated fabric. Coated fabric properties shall be tested in accordance with test methods in table I. Unless otherwise specified (see 6.2), all samples shall be taken from the coated fabric used in the construction of the tank wall. Nonconformance to 3.6.6 and table I shall constitute failure of this test.

4.5.1.11 Puncture resistance. FED-STD-191, method 5120 applies except that the ring clamp mechanism shall have an internal diameter of 3.00 inch, and the ball shall be replaced by a piercing instrument shaped like a flared, flat-tip screwdriver, having a width of 0.312, ±0.010 inch, and a thickness of 0.031, ±0.004 inch, at the extreme tip. The piercing tip edges shall be round to a 0.010 inch radius. The piercing instrument shall be oriented to intercept the warp and fill threads at an angle of approximately 45 degrees. The average of three test specimens shall be reported. Nonconformance to 3.6.6 and table I shall constitute failure of this test.

4.5.1.12 Low temperature crease resistance. Fold two specimens, each 8 inches square, in half in each direction so that a folded corner occurs in the center of each specimen. Place each folded specimen under a 4-pound load and condition at -28 ±2 °F, for 46 hours. At the end of the conditioning period, unfold the specimens after raising the temperature to -25 ±2 °F and examine visually. Signs of cracking, peeling or delamination of any coating material shall constitute failure of this test.

4.5.1.13 Blocking. Place two coated fabric specimens 6 inches by 1 inch in an oven on a smooth surface in such a manner that the ends are overlapped 1 inch. Place 4-pound weight directly on the overlapped area. After conditioning at a temperature of 160 ±2 °F, for 4 hours, remove the weight and

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take the specimens from the oven and condition for 1 hour at 73 ±5 °F. Attach one end of the specimen in a suitable clamping device allowing the free end to hang down. Suspend a 4-ounce weight from the free end of the specimens. Inability of the strips to separate within 5 seconds under the 4-ounce load shall constitute failure of this test.

4.5.1.14 Coating adhesion. Samples of coated fabric shall be bonded face-to-face to provide specimens for determining adhesion between the cloth and exterior coating(s), between the cloth and interior coating(s), between laminations of interior coating(s) and barrier (if used), and between laminations of exterior coating(s). In forming this bond the specimens shall be subjected to no heat or pressure other than that normally encountered in curing the coated fabric, except for minimal pressure necessary to insure contact while the bond is setting.

4.5.1.14.1 Test procedure. The adhesion shall be determined in accordance with ASTM D 413, machine method, except that the specimens shall be 2 inches wide. The specimens shall be of sufficient length to conduct adhesion tests for both initial values and after water immersions. The adhesive results obtained from each immersed specimen shall be compared with the initial adhesion of the same specimen to determine percentage of adhesion retained. The reported adhesion and percent retention shall be the average of not less than three specimens. Attempts shall be made to cut the coating back to the base cloth and to determine the adhesion value at the coating-to-cloth interface. However, if a specimen separates at a plane other than the bond of the coating to cloth (such as between layers of coating materials or between barrier film and coating) the adhesion value and the plane of failure shall be recorded and reported. Immersed specimens shall be conditioned in the test fluid at 73 ±5 °F, for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the conditioned fluid. Immersion of specimens shall be in accordance with ASTM D 471. Nonconformance to 3.6.6 and table I shall constitute failure of this test. Any obvious bond failure evident after immersion but before stressing, even if the plane of failure is not sandwiched between the layers of nylon fabric, shall constitute failure of this test.

4.5.1.15 Seam tests. The bonding together of any two or more pieces of coated fabric (such as lap joints, butt joints, and closures, chafing patches, coated fabric flanges of fittings, etc.) shall be considered as seams and shall be subjected to all seam tests specified herein, except chafing patches which will only be tested for peel. The average breaking strength of five specimens for each type seam for each test shall be reported for conformance to table II. Breaking strength specimens shall be 2 inches wide (parallel to the seam) and shall extend (perpendi-cular to the seam) 3 inches beyond both edges of the seam. No part of the test specimens shall be coated or covered during the water immersion periods. Specimens shall be cooled in the immersion fluids at 73 ±5 °F, for 30 to 90 minutes before testing. Testing of immersed specimens shall be completed within 3 minutes after removal from the immersion fluids. The average peel adhesion strength of three specimens for each type seam shall be reported for conformance to table II. Peel adhesion

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specimens shall be of sufficient length to determine both the initial and after water adhesion values on the same specimen. If seam construction involves the use of binding thread, then the peel specimens shall be prepared with threads removed. Nonconformance to 3.7.1 and table II shall constitute failure of this test.

4.5.1.16 Dead load shear resistance. The test specimens shall be 1.0 inch, ±0.020-inch wide, (parallel to the seam) and shall extend a minimum of 3 inches on each side of the seam. One index mark shall be scribed on each side of the seam to facilitate observation and measurements of slippage. Each specimen shall be subjected to a constant (dead load) tension force of 50 pounds, ±1/2 pound, at 160 ±2 °F. After 8 hours examine each specimen while still under tension for signs of slippage or separation. Three specimens shall be tested for each determination. Slippage, by any specimen, greater than specified in 3.7.1 and table II shall constitute failure of this test.

4.5.1.17 Strength of bonded reinforcing pad. Specimens shall be prepared by cutting through the assembly at an equal distance between screw holes and through center of screw holes so that wedge shape sections are obtained. These sections shall include at least two inches of the coated fabric beyond the edge of the reinforcing pad.

4.5.1.17.1 Initial bond strength. The coated fabric bonded to the reinforcing pad shall be fastened together in one jaw of the test machine so that the jaw will be at least 1.0 inch from the nearest part of the reinforcing pad. The reinforcing pad ring section shall be secured in the other jaw of the test machine and this jaw shall clamp only the ring and reinforcing pad. The jaws shall be separated at a rate of 0.5 inch per minute at 73 ±5 °F. The average of three test specimens shall be recorded as initial bond strength in pounds per inch of width. To determine this value the sample width shall be the cord length at the point of failure. Nonconformance to 3.7.3.1 and table III shall constitute failure of this test.

4.5.1.17.2 Bond strength after fluid immersion. Three test specimens shall be immersed for the appropriate durations in each test fluid specified in table III. No part of the specimens shall be covered or coated during immersion. The test specimens shall be cooled in the immersion fluid to 73 ±5 °F, for up to 60 minutes. The specimens shall be removed from the test fluid, one at a time and tested as in 4.5.1.17.1. Each test shall be completed within 3 minutes after removal from the test fluid. The average of three tests for each fluid shall be reported as bond strength after immersion in pounds per inch of width. Nonconformance to 3.7.3.1 and table III shall constitute failure of this test.

4.5.1.17.3 Dead load shear resistance reinforcing pad to fabric bond. Specimens shall be prepared by cutting through the reinforcing pad where it is bonded to the tank floor so as to produce 1-inch wide parallel specimens. These sections shall include at least 3 inches of coated fabric beyond the edge of the reinforcing pad. Three specimens shall be clamped as in

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4.5.1.17.1 and subjected to a constant (dead load) tension force of 50 pounds at 160 ±2 °F. At the end of 8 hours, the specimens shall be examined for slippage or separation while under tension. Nonconformance to 3.7.3.1 and table III shall constitute failure of this test.

4.5.1.18 Peel adhesion of reinforced pad to coated fabric. Specimens shall be prepared by cutting a 2-inch wide strip through the ring assembly extending into the coated fabric. These specimens shall include at least two inches of the coated fabric beyond the edge of the reinforcing pad.

4.5.1.18.1 Test procedures. Specimens shall be tested as per ASTM D 413, machine method, strip specimen type A. Separate single specimens shall be used to determine initial peel strength and peel strength after fluid immersion. The same identical specimens shall be used when computing the percentage of initial adhesion retained. Nonconformance to 3.7.3.1 and table III shall constitute failure of this test.

4.5.1.19 Stability. The tank shall be deployed on a 10 percent (minimum) slope and filled to rated capacity with water. Nonconformance to 3.19 shall constitute failure of this test.

4.5.1.20 Taste and odor. Specimens shall be tested in accordance with APHA Standards and Methods for Examination of Water and Wastewater, parts for Taste (211B) and Odor (207). The water samples for taste and odor testing shall be drawn from the water in the tank at the end of the water storage after repair test (see 4.5.1.5.2). Sampling and storage procedures shall be as specified in APHA 207/211B. The test panel shall include 10 persons, at least one of which shall be a Government represent-ative. The report for odor shall include the computed Threshold Odor Number as the geometric mean of the individual thresholds of each panel member. The report for taste shall include the mean and standard deviation of all ratings given for each sample. Nonconformance to 3.10.1 shall be cause for rejection.

4.5.1.21 Ozone resistance. Test according to ASTM D 1149 specimen B. Test specimen shall be conditioned for 7 days at 104 ±4 °F having a partial pressure of ozone of 50 milipascals. Nonconformance to table I shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 First article packaging inspection. When a first article pack is required (see 5.1 and 6.2), the first article packaging shall be examined for the defects listed in table VI. The presence of one or more defects shall be cause for rejection.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection, a completed tank, with all accessories, prepared for shipment shall be considered a unit of product.

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4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105. Sample size shall be determined by table I and table IIa, inspection level 1, of MIL-STD-105. A lot shall be accepted when zero defects are found, and rejected when one defect is found.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the defects listed in table VI.

TABLE VI. Packaging examination schedule.

Examination number	Examination description	Levels		
		A	B	C
132.	Preservation not as specified.	5.2		
133.	Preservatives not compatible with potable water.	5.2		
134.	Repair items not preserved as specified.	5.2.1		
135.	Fittings not preserved as specified.	5.2.2		
136.	Inflation bellows not preserved as specified.	5.2.3		
137.	Protective cover, float and float retaining net not preserved as specified.	5.2.4		
138.	Tanks not dusted as specified.	5.2.5		
139.	Tanks not folded or rolled to the desired size as specified.	5.2.5		
140.	Ground cloth not dusted as specified.	5.2.6		
141.	Ground cloth not secured with web ties as specified.	5.2.6		
142.	Securing means for bundle not as specified.	5.2.6.1		
143.	Lumber and plywood not treated with wood preservative and dried as specified.	5.3.1.1		
144.	Carriage bolts, lag screws, nuts and washers not as specified.	5.3.1.2	5.3.2	
145.	Plywood not as specified.	5.3.1.2.1	5.3.2	
146.	Cleats not as specified.	5.3.1.2.2	5.3.2	
147.	Skids not as specified.	5.3.1.2.3	5.3.2	
148.	Fabrication of shipping container panels not as specified.	5.3.1.2.4	5.3.2	
149.	Assembly not as specified.	5.3.1.2.5	5.3.2	
150.	Anti-abrasion provisions not as specified.	5.3.1.3	5.3.2	5.3.3
151.	Marking not as specified.	5.4	5.4	5.4
152.	Special marking for handling preservative treated wood not as specified.	5.4.1		

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5. PACKAGING

5.1 First article packing. Unless otherwise specified (see 6.2), the contractor shall present a first article pack for examination within the time frame specified (see 6.2), to prove prior to the start of production packaging, that the applied preservation, packing and marking comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.2). The first article pack may utilize either the first article model or a production model. When the first article model is utilized, and the Government requests a comparison between the first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 Preservation. Each complete tank shall be preserved to satisfy level A preservation requirements in the following manner. All ferrous type metal surfaces requiring a contact preservative in accordance with MIL-P-116 shall be coated with P-14 preservative in accordance with MIL-P-116. Preservative materials and their application shall not adversely affect the potability of the water to be stored in the tanks. The complete tank, when preserved as specified, shall be suitable for placement in the valise specified in 3.9.

5.2.1 Repair items (see 3.11).

5.2.1.1 Repair kit. Each component of the repair kit shall be preserved in a bag in accordance with the level A requirements of MIL-R-52255. The bags shall be sealed and marked in accordance with the documents referenced therein.

5.2.1.2 Other repair items. The remaining repair items shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3 and items of like description for each tank shall be preserved together in such a manner as to prevent damage or deformation.

5.2.2 Fittings. The cap for each fitting shall be installed on the fitting. The fitting shall then be covered with cushioning material conforming to PPP-C-843, type II, class B, or PPP-C-795 or PPP-C-1797. The cushioning material shall be secured in place on the tank with tape conforming to MIL-T-22085, type II, PPP-T-60, type III or IV, or PPP-T-97, type IV.

5.2.3 Inflation bellows (see 3.10). The inflation bellows shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3. Alternatively, the bellows may be preserved in accordance with the level A requirements of its applicable specification.

5.2.4 Protective cover. The protective cover, including the float, (see 3.13) shall be preserved in accordance with the level A requirements of their applicable specification(s). Alternatively, the cover, float, and float retaining net may be laid flat and folded or rolled with the tank when the tank is being folded or rolled as specified in 5.2.5.

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5.2.5 Tank. Each tank shall be coated with an antisticking compound to prevent adhesion to itself. The inflation collar shall be coated on the inside and outside. The antisticking compound shall not adversely affect potable water. The tank shall be laid flat, completely collapsed, and then folded or rolled to form a neat compact bundle.

5.2.6 Ground cloth. Each ground cloth shall be coated with an antisticking compound to prevent adhesion to itself. The cloth shall be laid flat and folded or rolled to form a neat compact bundle capable of being positioned within the storage/carrying case (see 3.9) with the tank and all other accessories. The ground cloth bundle shall be secured as a tight bundle with web ties.

5.2.6.1 Securing means. The means of securing the bundle shall not subject the item to the hazards of any sharp metal objects utilized either as a component of the securing device or as a tool for releasing the means.

5.3 Packing. Each complete tank, preserved as specified in 5.2, shall be packed in accordance with the requirement for level A, level B or level C as specified (see 6.2).

5.3.1 Level A. When level A packing is specified (see 6.2) the following special considerations shall be afforded this product because of an anticipated extended period of storage under favorable environmental conditions.

5.3.1.1 Wood preservation (see 6.11). All lumber and plywood used in the construction of the exterior crates/boxes shall be treated with wood preservative in conformance with composition C or D of TT-W-572. The following additional conditions must be satisfied:

a. Treatment with composition C shall result in a minimum retention of 0.040 pounds of copper per cubic foot of lumber/plywood as defined in the applicable standard(s) referenced in C1, C2, and C9 of the AWPA.
b. After treatment with either composition, the material shall be dried to a moisture content of 19 percent or less.

5.3.1.2 Exterior containers/shipping containers. Exterior/shipping containers shall be snug-fitting boxes conforming to PPP-B-601, overseas type, style A, type 3 load, except for the following requirements. Carriage bolts, lag screws/bolts, nuts and washers shall conform to the following:

CARRIAGE BOLTS	-	FF-B-584, type I, class 1, style A
LAG SCREWS/BOLTS	-	FF-B-561, type I, grade B
NUTS	-	FF-N-836, type I or II, style I or 4
WASHERS	-	FF-W-92, type A, grade I, class A

5.3.1.2.1 Plywood. All plywood shall be in accordance with AA-55057. The plywood shall be interior type, bonded with exterior (waterproof) glue, grade CD.

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5.3.1.2.2 Cleats. All lumber for the cleats shall comply with the quality requirements of MIL-STD-731 for class 2 - Structural (members with moderate stress), except that wane is not acceptable. The cleat sizes shall be nominal 1 inch x 4 inches and nominal 2 inches x 4 inches with minimum dimensions for each as shown therein.

5.3.1.2.3 Skids. All lumber for the skids shall comply with the quality requirements of MIL-STD-731 for class 2 - Structural (members with moderate stress), except that wane is not acceptable. Skids shall be nominal 4 inches x 4 inches. Group I wood species shall not be used for the skids.

5.3.1.2.4 Fabrication. Each panel of the shipping container shall be double sheathed with plywood. The inside plywood shall be 0.25 inches thick and the outside plywood sheathing shall be 0.50 inches thick. The outside sheathing shall be attached to the applicable cleats with the fasteners driven through the C side of the plywood and clinched on the exposed side of the cleat. The inside sheathing shall then be attached to the panel with the fasteners driven through the wood parts in the sequence of the C side of the 0.25-inch plywood, to cleat, to 0.50-inch plywood. The sizes of the fasteners, the positioning of the fasteners and the clinching of the fasteners shall be as specified in PPP-B-601; however, care shall be exercised so that the positioning of such fasteners does not interfere with the lag screws/bolts and carriage bolts required to assemble the top, sides, ends and bottom panels together. Clinching of the fasteners shall not be permitted on any inside surface of the container panels.

5.3.1.2.4.1 Side panels. The side panels shall utilize nominal 1-inch x 4-inch lumber for the cleats (see 5.3.1.2.2). The positioning of the edge cleats and intermediate cleats when required, shall be as specified for the sides in PPP-B-601. The size of the side panel shall be such that its length (the longest dimension) is the same as the outside length of the box. The width of the side panel shall be such that it is the same as the inside depth of the box plus the thickness of the double sheathed bottom panel, (see figure 3).

5.3.1.2.4.2 Top panel. The top panel shall utilize nominal 2-inch x 4-inch lumber for the cleats (see 5.3.1.2.2). The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the top in PPP-B-601. The size of the top panel shall be as specified therein for style A; that is, the length and width of the panel shall be the same as the outside length and outside width of the box. (see figures 3 and 4).

5.3.1.2.4.3 End panels. The end panels shall utilize nominal 2-inch x 4-inch lumber for the cleats (see 5.3.1.2.2). The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the ends in PPP-B-601. The size of the end panel shall be the same as specified therein for style A; that is, the length and width shall be the same as the inside width and inside depth of the box, (see figure 3).

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5.3.1.2.4.4 Bottom panel. The bottom panel shall utilize nominal 2-inch x 4-inch lumber for the cleats (see 5.3.1.2.2). The positioning of the edge cleats, and intermediate cleats when required, shall be as specified for the bottom in PPP-B-601 except that the thru edge cleats shall run parallel with the box length and such cleats shall be the same length as the outside length of the box. The size of the bottom panel shall be as specified therein for style A except that the width of the bottom panel shall be the same as the inside width of the box, (see figure 3).

5.3.1.2.4.5 Skids. Skids (see 5.3.1.2.3), shall be applied to the bottom panel of each box, running parallel with and extending the full outside width of the box. A bevel 45 degrees (±5 degrees) shall be applied to the bottom third portion of the skid ends. The skids shall be set back from the end of the box a distance of not less than 3 inches or more than 1/6 of the box length. Each box shall be provided with a minimum of 2 skids with additional skids being provided when the distance between the inside edges of the skids exceed 48 inches. The positioning of the additional skids shall divide the area between the ends skids into units of equal space. Additional intermediate cleats shall be incorporated into the bottom panel at time of fabrica-tion to assist in the attachment of the skids and to offer full support to the skids. Each skid shall be fastened to the bottom panel with 0.375 inch diameter carriage bolts that are 5.25 inches long (+0.00 inches, -0.25 inches). The bottom panel shall be positioned so that it is set back from the ends of the skids a distance that is equal to the thickness of the box side panels. The holes for the carriage bolts shall be the exact diameter of the bolt. Each hole shall be counterbored on the underside of the skid so that 3 full threads (+3,-0) extend beyond the washer and nut when the nut has been properly tightened. The nut shall be tightened so that the bolt head is drawn into the plywood and cleat to prevent the bolt head from becoming a snagging hazard on the inside of the box; however, no portion shall extend beyond the bottom of the skid when tightened. After tightening the nuts, the threads of the bolt extending beyond the nut shall be painted with a suitable metal primer or similar material. The bolts shall be centered at 6.50 inches (±0.25 inches) from each end of the skid with additional bolts positioned in a staggered pattern, as practicable, so that the distance between their center-lines shall be no more than 16 inches (±0.25 inches) and no less than 8 inches (±0.25 inches), (see figure 6). Each skid shall be slotted/notched sufficiently to provide clearance for strapping. Figures 4, 5 and 6 give details of skid application.

5.3.1.2.5 Assembly. The box panels shall be joined together with lag screws/bolts and flat washers. Entry holes and lead holes shall be provided in each box panel for proper assembly as follows:

5.3.1.2.5.1 Entry holes. Entry holes, the same size as the lag screws, shall be provided through each applicable flat panel surface. The holes shall be set back from the edges of that flat surface to provide the edge distance necessary for the lag screws to be centered on the nominal 1-inch, or as applicable the nominal 2-inch, cleat thickness of the joining panel. The entry holes for the 0.25 inch diameter lag screw for joining a panel to a

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nominal 1-inch thick cleat shall be centered at 0.875 inch from the edge of the flat panel. The entry holes for the 0.375-inch diameter lag screw for joining a panel to a nominal 2-inch thick cleat shall be centered at 1.25 inches from the edge of the flat surface (see figures 3, 4, 5 and 6).

5.3.1.2.5.2 Lead holes. Lead holes for the threaded portion of the lag screw shall be provided in each applicable panel edge. The holes shall be drilled only to the depth necessary to accommodate the penetration of the threaded portion of the lag screw. The lead holes for the 0.25-inch diameter lag screw shall be 0.1875-inch in diameter. The lead holes for the 0.375 inch-diameter lag screws shall be 0.25 inches in diameter. The lead holes shall provide the same edge distance as specified in 5.3.1.2.5.1 for the entry holes.

5.3.1.2.5.3 Lag screw application. The lag screws shall be turned in their holes the full length of the screw. They shall not be driven in by hammer or by any other similar means. If for any reason the thread in the wood is stripped when the lag screws are placed, the lag screw shall be removed and placed in a new hole near the old position. One flat washer shall be used under the head of each screw.

5.3.1.2.5.4 Fastening bottom assembly to end panels. The bottom assembly shall be fastened to each end panel with a minimum of 4 lag screws 0.375 inch in diameter by 5 inches in length. The positioning of each screw shall be as shown in figure 3.

5.3.1.2.5.5 Fastening side panels to end panels. The side panels shall be fastened to the end panels with 0.375 inch diameter lag screws that are 4 inches long. The screws shall be positioned as shown in figure 4.

5.3.1.2.5.6 Fastening of side panels to bottom assembly. The side panels shall be fastened to the bottom assembly with 0.375 inch diameter lag screws that are 4 inches long. The screws shall be positioned as shown in figure 4.

5.3.1.2.5.7 Fastening top panel to side panels. The top panel shall be fastened to the side panels with 0.25-inch diameter lag screws that are 5 inches long. The screws shall be positioned as shown in figure 5.

5.3.1.2.5.8 Fastening top panel to end panels. The top panel shall be fastened to the end panels with 0.375 inch diameter lag screws that are 5 inches long. The screws shall be positioned as shown in figure 5.

5.3.1.2.5.9 Strapping. Strapping of the assembled box shall be as specified in the appendix to PPP-B-601 except that the strapping shall be flat and the finish shall be "B".

5.3.1.3 Anti-abrasion provisions. To prevent abrasion, all interior surfaces of the shipping container that will contact the tank and tank components, shall be lined with a layer of scrap tank fabric, cushioning material conforming to PPP-C-1797, or polyethylene film conforming to L-P-378,

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type I, class 1, with a minimum thickness of 6 mils. The material shall be applied in such a manner that it does not interfere with the removal of the container top, sides and ends as a joined one piece assembly or as individual panels. The means of attaching the material shall not create a hazard within the container that could possibly damage the contents. As an alternative, the complete tank, preserved as required in 5.2.1 through 5.2.6.1, shall be individually loosely wrapped with the anti-abrasion material. The manner of wrapping shall provide a minimum of one layer of material around the tank to prevent direct contact with other components and the inside surfaces of the shipping containers.

5.3.2 Level B. Each complete tank, preserved as specified in 5.2, shall be packed for level B in the same manner as specified for level A in 5.3.1 except that the lumber and plywood preservation requirements of 5.3.1.1 shall be omitted.

5.3.3 Level C. Each complete tank, preserved as specified in 5.2, shall be individually packed in a close-fitting box conforming to PPP-B-601, domestic type, style A or PPP-B-621, class 1, style 2. Beveled skids shall be applied to the box bottoms in accordance with the applicable box specifications. In order to prevent abrasion, the box shall be loosely lined with polyethylene film conforming to L-P-378, type I, class 1, having a minimum thickness of 0.006 inches. Box closure and strapping shall be in accordance with the box specification or the appendix thereto except that the strapping shall be flat and the finish may be A.

5.4 Marking. In addition to any special or identification marking required by the contract or purchase order (see 6.2), each complete tank shall be marked in accordance with MIL-STD-129. Other special markings are as follows:

5.4.1 Marking for handling of treated wood. Each box constructed of treated lumber and plywood, shall, as applicable, be marked on an exterior surface in a prominent position with instructions advising involved personnel of any precautions to be exercised in the handling of the treated lumber. Such precautions shall include recommended limitations in the use and disposal of the treated lumber and shall be in the format developed by the treated wood industry's Consumer Awareness Program (CAP) as approved by the Environmental Protection Agency (EPA).

5.4.1.1 Methods of marking. The instructions required in 5.4.1 shall be applied to the shipping container in the manner specified in MIL-STD-129 for stenciling or labeling.

5.4.1.2 Alternative method. As an alternate, such instructions may be as preprinted matter to be included with the packing list in the exterior packing list envelope. When the instructions are placed in the envelope, a suitable area immediately adjacent to the envelope shall be marked with the words "INSTRUCTIONS FOR HANDLING TREATED WOOD INSIDE PACKING LIST ENVELOPE", with an arrow pointing to that envelope.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tank is intended for use as a potable water storage container when quick storage facilities are needed and where permanent potable water storage facilities are not available, or when the storage of potable water is only on a temporary basis.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article tests, the contracting officer should specify which examination and test shall be conducted by the Government and which examination and test shall be conducted by the contractor (see 3.2 and 4.3).
d. Time frame for availability of sample selection plan (see 3.3).
e. Time frame for availability of process data (see 3.5).
f. Time frame for availability of the rework and repair procedure (see 3.20).
g. Specify color (see 3.15).
h. Specify when samples shall not be taken from the coated fabric used in the construction of the tank wall (see 4.5.1.10).
i. When a first article pack is not required (see 5.1).
j. Time frame for presentation of first article pack, when required (see 5.1).
k. Level of packing required (see 5.3).
l. Any special marking (see 5.4).

6.3 Sample selection plan. The sample selection plan (see 3.3 and 6.2d) and process data (see 3.5 and 6.2e) require Government approval prior to construction of the first article tank.

6.4 First article. When a first article inspection is required, the item(s) should be initial production models. The first article should consist of two units, one of which will be used in destructive testing. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Solicitations should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid or proposal that prior Government approval

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is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack and mark the tanks in accordance with this specification.

6.6 Process data. At the time of first article tank fabrication, the contractor shall make available to the Government documented work instructions and processing and fabricating procedure to be used by the contractor to assure that each production tank is a duplicate of the first article tank. Change of materials, components, procedures, or sources of supply from those initially selected shall be subject to approval of the contracting officer. Inspection will be performed in accordance with the requirements of sections 3 and 4 and will include the identification of component lots into serially identified end items.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of tanks.

6.8 Subject term (key word) listing.

Onion tank
Potable water

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.10 Level of preservation. For the purpose of marking, the preservation specified in 5.2 shall be designated as level A.

6.11 Wood preservatives (see 5.3.1.1). The intent of the wood preservative application specified in 5.3.1.1 is to prolong the serviceability of the wooden shipping containers. Compositions C and D of TT-W-572 can offer reliable protection for periods up to 30 years against decay and wood destroying organisms provided such treated containers are not stored in direct contact with the ground.

6.12 Material safety data sheets. Contracting officers shall identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313, appendix B (see 3.11.1).

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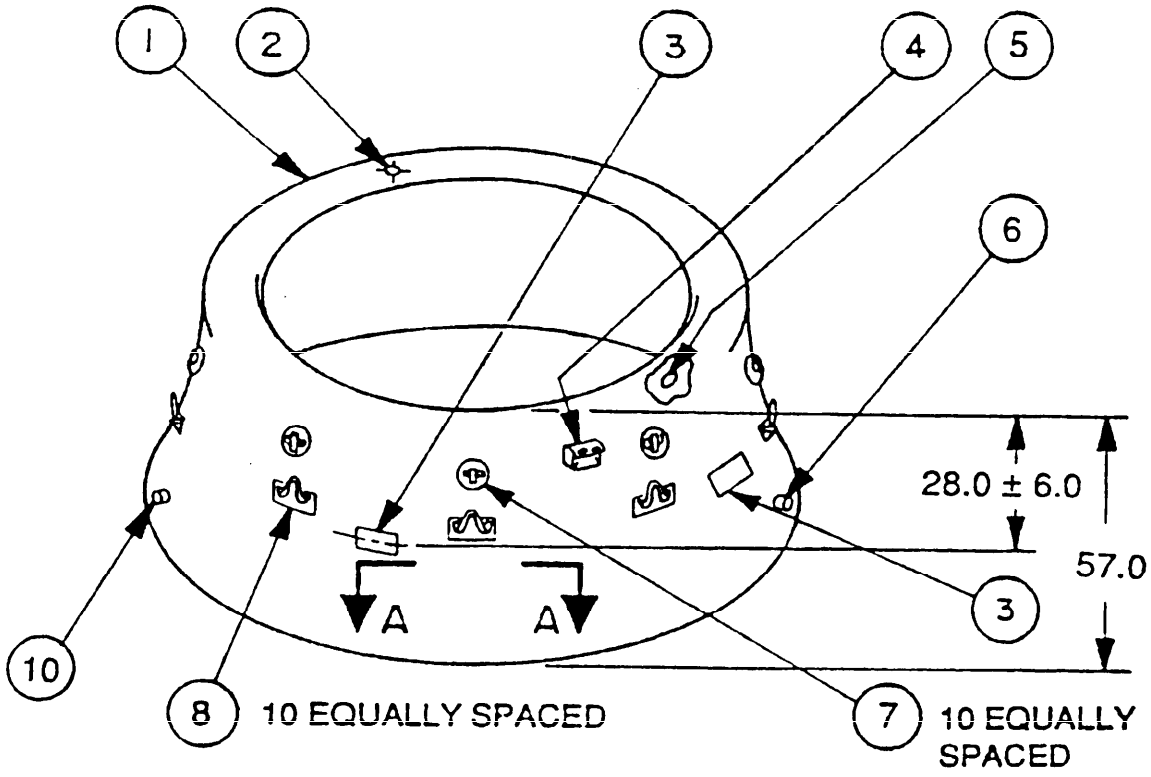
Custodian:
Army - ME
Navy - YD

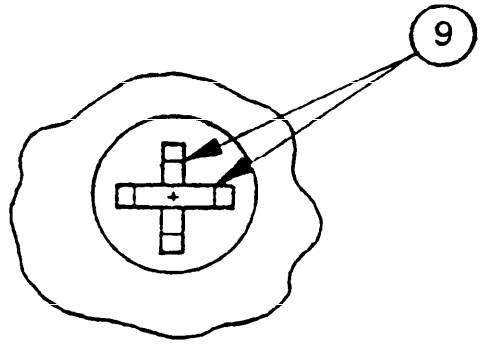
Review:
Navy - YD

Preparing activity:
Army - ME

Project 5430-0220

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VIEW A-A

ITEM	DESCRIPTION	QTY
1	INFLATION COLLAR	1
2	INFLATE / DEFLATE PORT	1
3	LABEL	2
4	REPAIR KITS	2
5	SCHRADER VALVE, AUTOMOTIVE TYPE	1
6	MALE FILLER/ DISCHARGE ASSEMBLY	1
7	TOGGLE	10
8	HANDLE	10
9	HANDLE, CENTER	2
10	FEMALE FILLER/ DISCHARGE ASSEMBLY	1

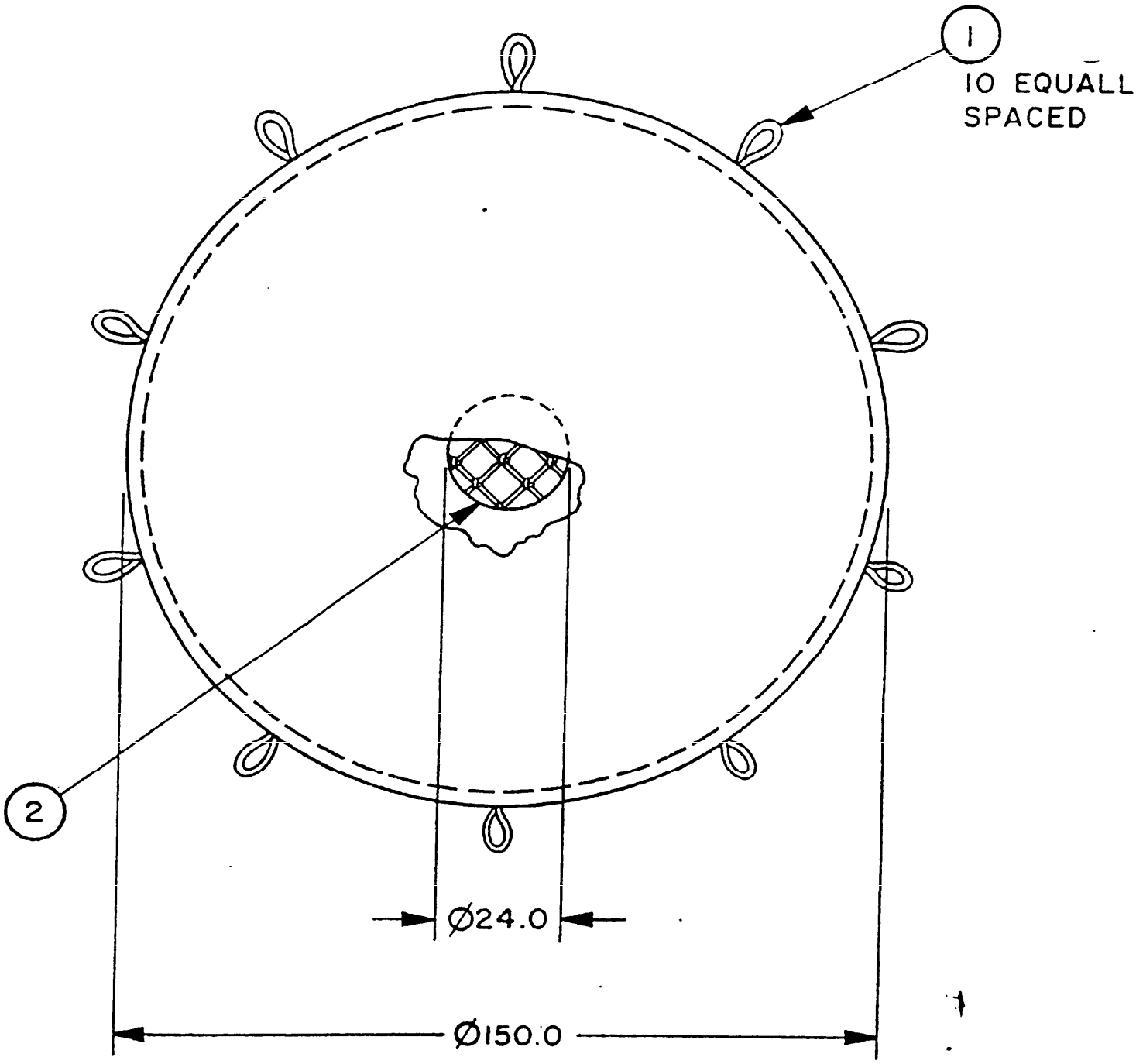
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. OPTIONAL DESIGN HANDLES MAY BE USED AS TOGGLES.

FIGURE 1. 3000 gal water storage tank.

X-4503C

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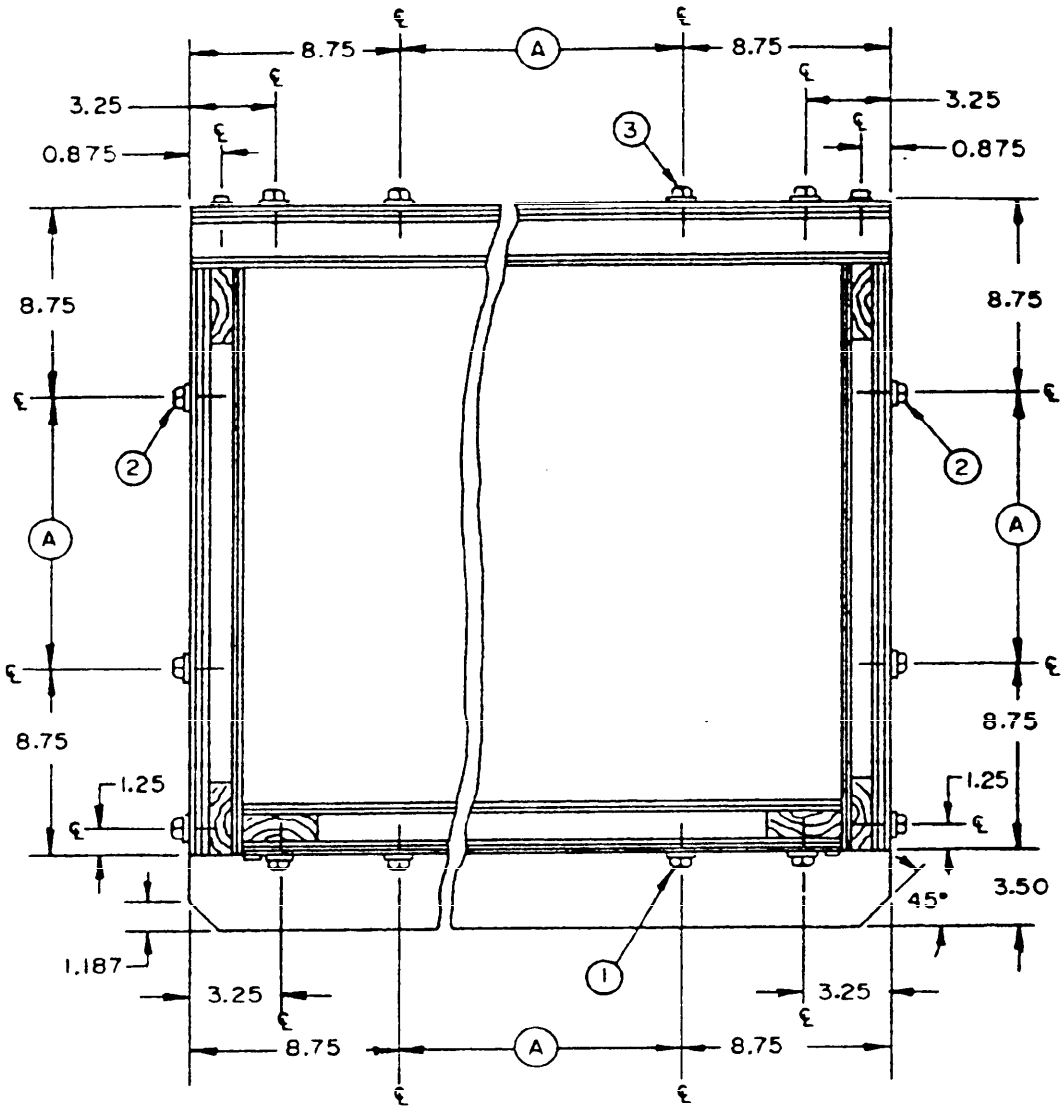
ITEM	DESCRIPTION	QTY
1	NYLON WEBBING LOOP	10
2	NET FOR BALL	1

NOTES:
1. ALL DIMENSIONS ARE IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ±3.0.
3. ITEM 2 IS AN OPTIONAL DESIGN.

FIGURE 2. Cover.

X-4504A

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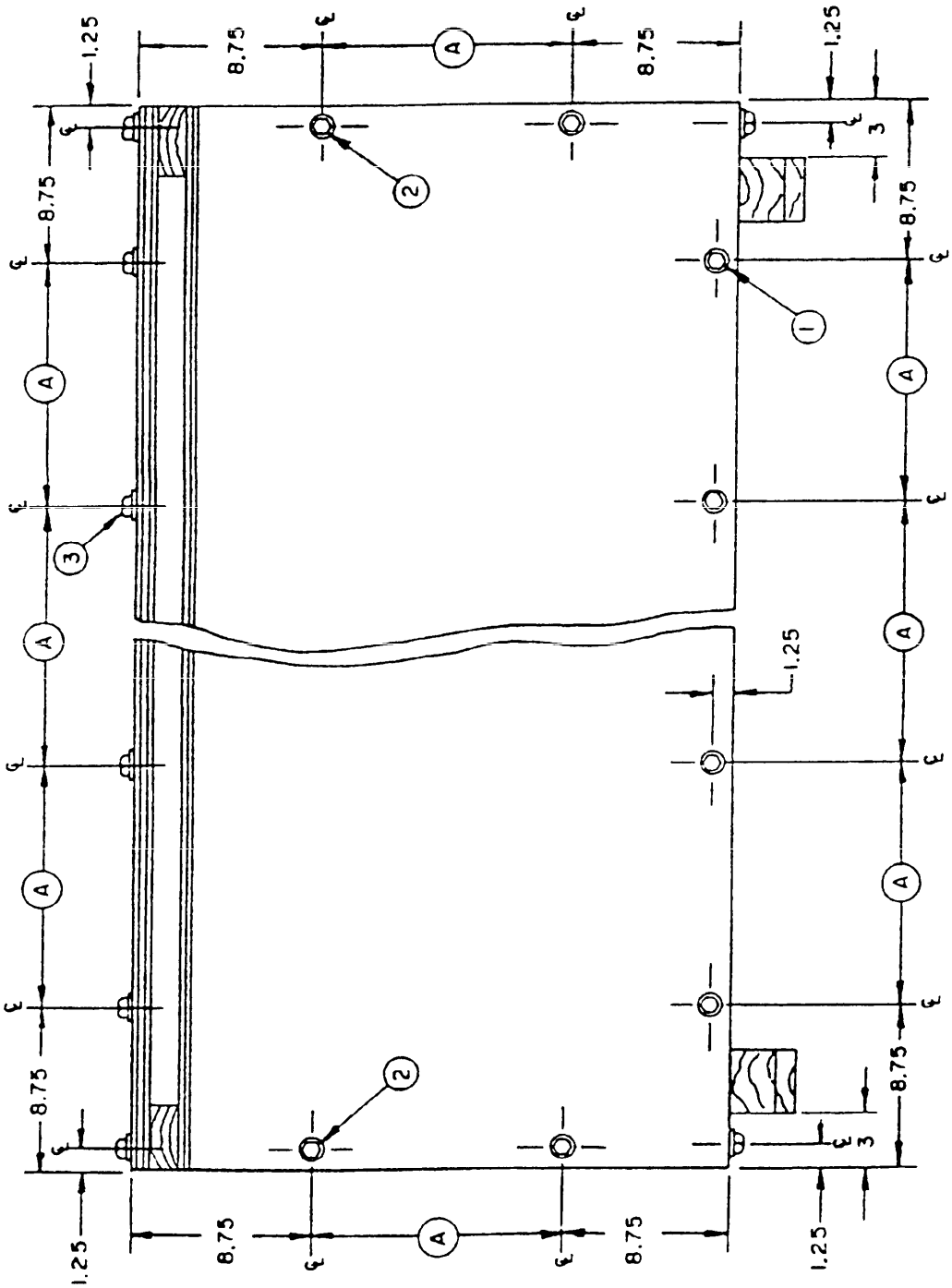


NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) BOTTOM ASSEMBLY TO END PANEL.
   - (2) SIDE PANEL TO END PANEL.
   - (3) TOP PANEL TO END PANEL.

FIGURE 3. End View of Box

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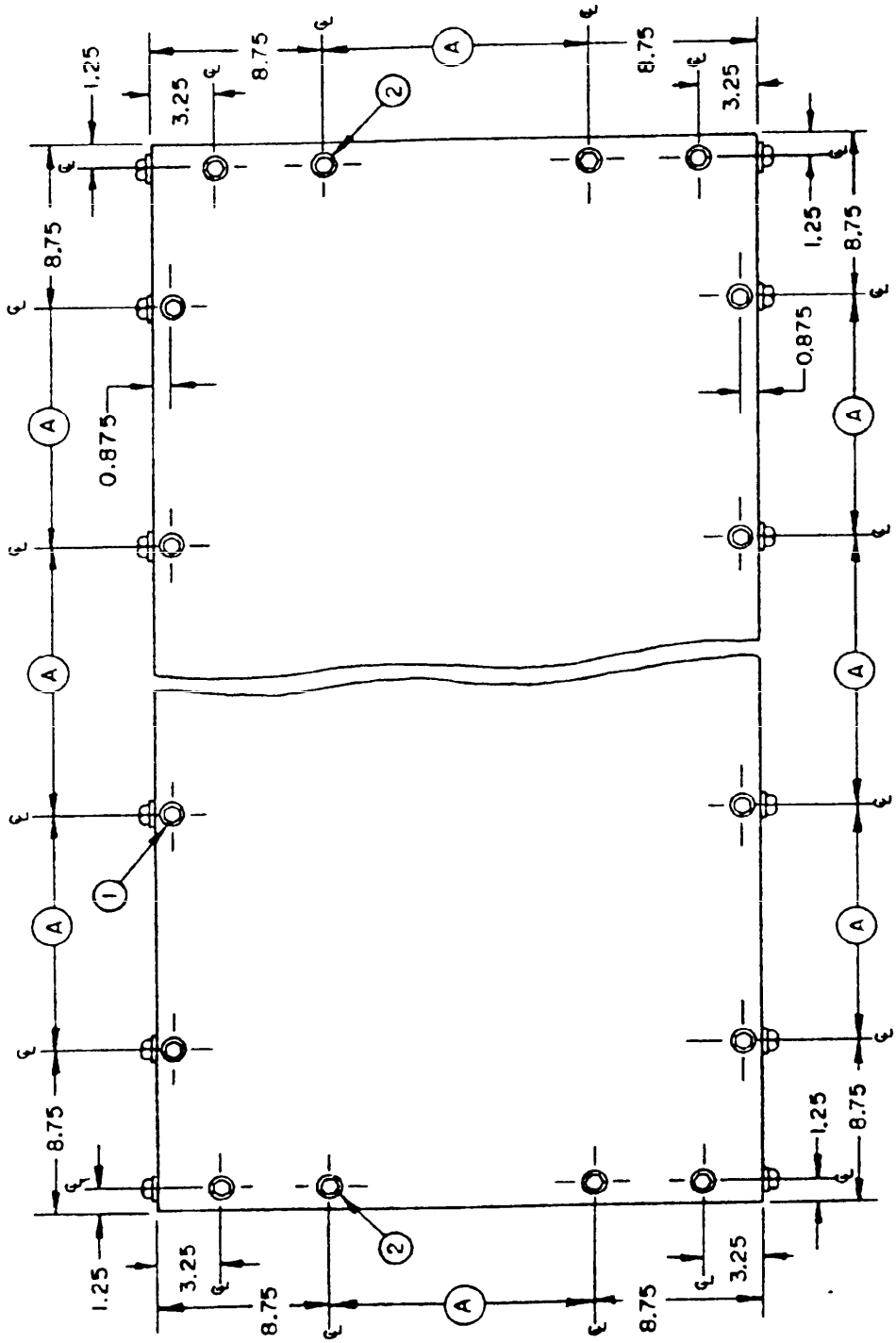
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) SIDE PANEL TO BOTTOM PANEL.
   - (2) SIDE PANEL TO END PANEL.
   - (3) TOP PANEL TO SIDE PANEL.

FIGURE 4. Side view of Box

X-4708

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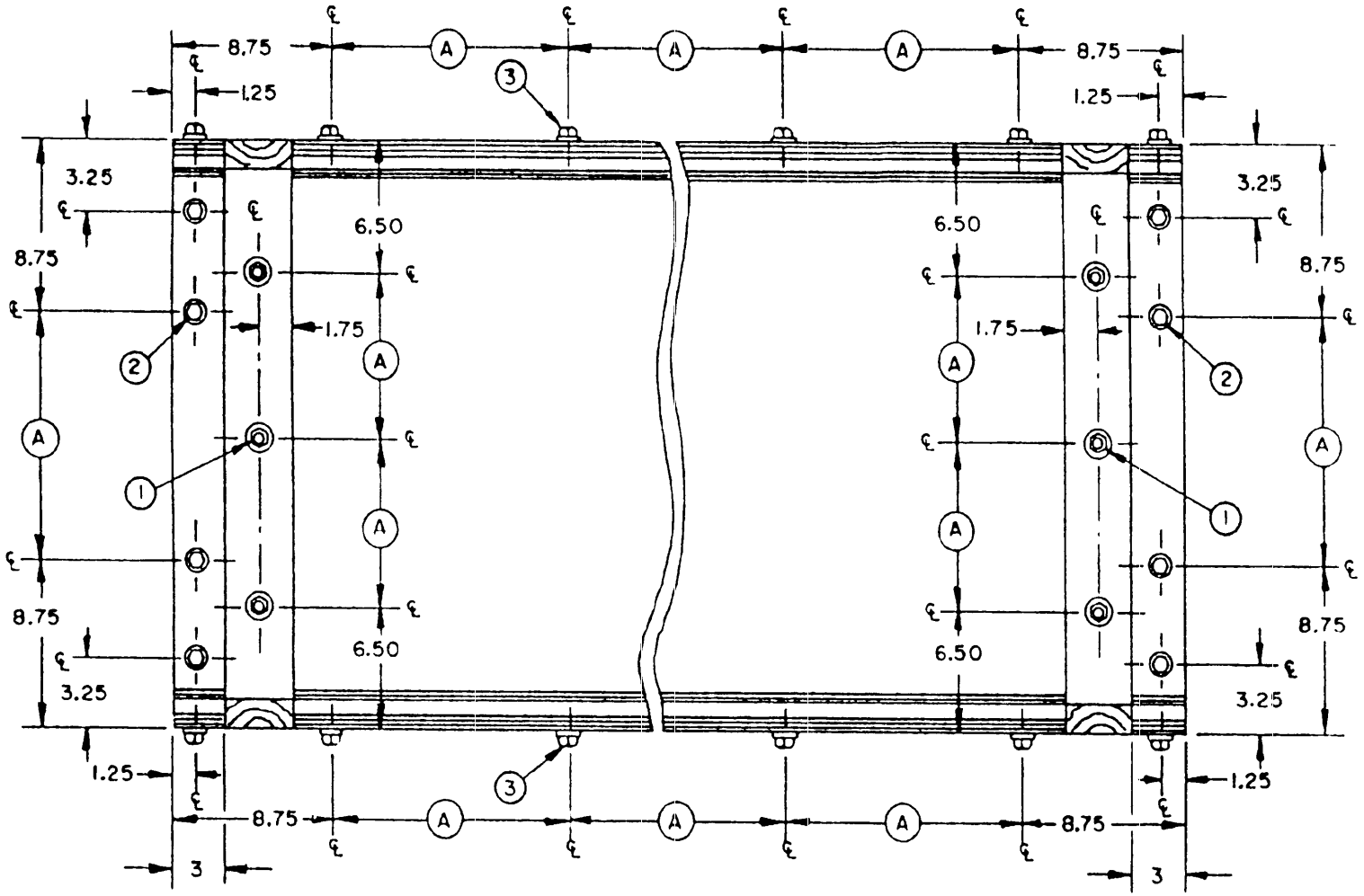
NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:
   - (1) TOP PANEL TO SIDE PANEL.
   - (2) TOP PANEL TO END PANEL.

FIGURE 5. Top View of Box

X-4709A

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NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. DIMENSIONS (A) BETWEEN INTERMEDIATE LAG SCREWS OR CARRIAGE BOLTS SHALL BE NOT LESS THAN 8 INCHES OR MORE THAN 16 INCHES.
3. FIGURE SHOWS FASTENING OF:

   (1) SKID TO BOTTOM PANEL.
   (2) BOTTOM PANEL TO END PANEL.
   (3) SIDE PANEL TO BOTTOM PANEL.

FIGURE 6. Bottom View of Box

X-4110A

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